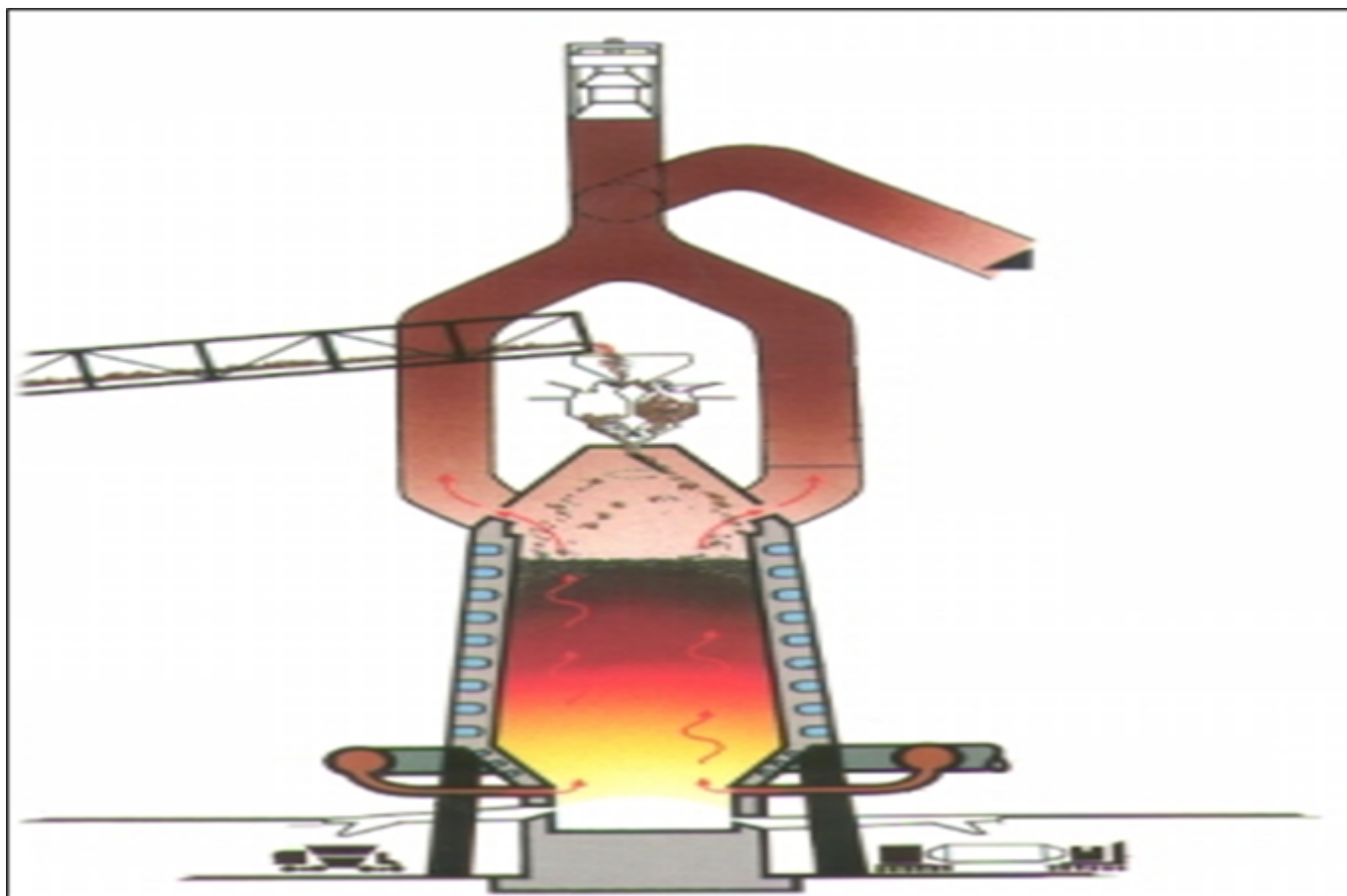


Qualification Pack



Operator - Green hydrogen injection (Blast Furnace)

QP Code: ISC/Q2301

Version: 1.0

NSQF Level: 4

Indian Iron and Steel Sector Skill Council || Plot no.B-7, Action Area-III, 5th Floor, Room no.509, Near
Hazra Kalibari, Opposite Uni World City, New Town, Kolkata, West Bengal
Pin-700156 || email:akash@iisssc.org



Qualification Pack

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Qualification Pack

ISC/Q2301: Operator - Green hydrogen injection (Blast Furnace)

Brief Job Description

The individual in this role performs operation for Hydrogen injection in Blast Furnace and its associated functions.

Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0009: Work effectively with others](#)
3. [ISC/N2301: Prepare for operation for Hydrogen injection in Blast Furnace](#)
4. [ISC/N2302: Perform operation for Hydrogen injection in Blast Furnace through Tuyeres](#)
5. [ISC/N1704: Apply Decarbonization practice in plant](#)
6. [ISC/N1703: Apply statistical process control \(SPC\) techniques](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Green Hydrogen
Occupation	Blast Furnace Operation
Country	India
NSQF Level	4
Credits	16



Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO-2015/3122.0200
Minimum Educational Qualification & Experience	12th Class (or equivalent Science (Physics/Chemistry)) with 6 Months of experience in relevant field OR 11th Class (Science (Physics/Chemistry)) with 1.5 years of experience of experience in relevant field OR 10th Class with 3 Years of experience of experience in relevant field OR 9th Class with 4.5 years of experience of experience in relevant field OR Certificate-NSQF (Level 3) with 3 Years of experience of experience in relevant field
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQF Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-04-IS-04890-2025-V1-ISSSC
NQR Version	1



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Use protective clothing/equipment for specific tasks and work conditions
Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks
Glasses/gloves/visors
Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator

Health and safety procedures

To be competent, the user/individual on the job must be able to:

- PC2.** State the name and location of people responsible for health and safety in the workplace
Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.
- PC3.** State the names and location of documents that refer to health and safety in the workplace
- PC4.** Identify job-site hazardous work and state possible causes of risk or accident in the workplace
Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.
- PC5.** Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role
Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors
- PC6.** State location of general health and safety equipment in the workplace



Qualification Pack

- PC7.** Inspect for faults, set up and safely use steps and ladders in general use
Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.
Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.
- PC8.** Work safely in and around trenches, elevated places and confined areas
- PC9.** Lift heavy objects safely using correct procedures
- PC10.** Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces
- PC11.** Identify common hazard signs displayed in various areas
- PC12.** Retrieve and/or point out documents that refer to health and safety in the workplace

Fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC13.** Use the various appropriate fire extinguishers on different types of fires correctly.
Fire extinguishers: Sand Water Foam Co2 Dry powder
Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents)
Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.
- PC14.** Demonstrate rescue techniques applied during fire hazard
- PC15.** Demonstrate good housekeeping in order to prevent fire hazards
- PC16.** Demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC17.** Demonstrate how to free a person from electrocution
- PC18.** Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC19.** Demonstrate basic techniques of bandaging
- PC20.** Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below :
Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.
- PC21.** Perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC22.** Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC23.** Demonstrate the artificial respiration and the CPR Process
- PC24.** Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work



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- PC25.** Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer
- PC26.** Demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace
- KU2.** State the names and location of documents that refer to health and safety in the workplace
- KU3.** Meaning of hazards and risks
- KU4.** Health and safety hazards commonly present in the work environment and related precautions
- KU5.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** Activities and causes of risk and accident
- KU7.** Methods of accident prevention
- KU8.** Safe working practices when working with tools and machines
- KU9.** Safe working practices while working at various hazardous sites
- KU10.** Where to find all the general health and safety equipment in the workplace
- KU11.** Various dangers associated with the use of electrical equipment
- KU12.** Preventative and remedial actions to be taken in the case of exposure to toxic materials. Exposure: ingested, contact with skin, inhaled Preventative action: ventilation, masks, protective clothing/equipment Remedial action: immediate first aid, report to supervisor Materials: solvents, flux, lead
- KU13.** Importance of using protective clothing/equipment while working
- KU14.** Precautionary activities to prevent the fire accident Activities and causes: Physical actions Reading Listening to and giving instructions Inattention Sickness and incapacity (e.g. Drunkenness) Health hazards (e.g. Untreated injuries and contagious illness)
- KU15.** Various causes of fire
- KU16.** Techniques of using the different fire extinguishers
- KU17.** Different methods of extinguishing fire
- KU18.** Rescue techniques applied during a fire hazard
- KU19.** Various types of safety signs and what they mean
- KU20.** Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU21.** Content of written accident report



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- KU22.** Potential injuries and ill health associated with incorrect manual handling
- KU23.** Safe lifting and carrying practices
- KU24.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU25.** Potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Give clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS16.** Report problems that you cannot resolve to appropriate authority
- GS17.** Identify cause and effect relations in their area of work
- GS18.** Use cause and effect relations to anticipate potential problems and their solution



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	5	5	-	-
PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visors Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator	5	5	-	-
<i>Health and safety procedures</i>	15	40	-	-
PC2. State the name and location of people responsible for health and safety in the workplace Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.	-	4	-	-
PC3. State the names and location of documents that refer to health and safety in the workplace	-	1	-	-
PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors	5	5	-	-
PC6. State location of general health and safety equipment in the workplace	-	5	-	-
PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc. Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.	-	5	-	-
PC8. Work safely in and around trenches, elevated places and confined areas	-	4	-	-
PC9. Lift heavy objects safely using correct procedures	-	4	-	-
PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces	-	1	-	-
PC11. Identify common hazard signs displayed in various areas	5	1	-	-
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	-	5	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fire safety procedures</i>	10	15	-	-
PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: Sand Water Foam Co2 Dry powder Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.	5	5	-	-
PC14. Demonstrate rescue techniques applied during fire hazard	5	5	-	-
PC15. Demonstrate good housekeeping in order to prevent fire hazards	-	1	-	-
PC16. Demonstrate the correct use of a fire extinguisher	-	4	-	-
<i>Emergencies, rescue and first-aid procedures</i>	15	45	-	-
PC17. Demonstrate how to free a person from electrocution	-	5	-	-
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	5	-	-
PC19. Demonstrate basic techniques of bandaging	-	5	-	-
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.	5	5	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments	-	5	-	-
PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	-	5	-	-
PC23. Demonstrate the artificial respiration and the CPR Process	-	5	-	-
PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work	-	4	-	-
PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer	5	5	-	-
PC26. Demonstrate correct method to move injured people and others during an emergency	-	1	-	-
NOS Total	45	105	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0009: Work effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace

Elements and Performance Criteria

Ensure appropriate communication with superiors, peers and others as applicable at work place

To be competent, the user/individual on the job must be able to:

- PC1.** Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt
- PC3.** Provide information to others clearly, at a pace and in a manner that helps them to understand

Demonstrate appropriate behaviour and etiquette at work place

To be competent, the user/individual on the job must be able to:

- PC4.** Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible
- PC5.** Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC6.** Display appropriate communication etiquette while working
- PC7.** Display active listening skills while interacting with others at work
- PC8.** Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
- PC9.** Demonstrate responsible and disciplined behaviours at the workplace
- PC10.** Escalate grievances and problems to

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** Reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** Relevant people and their responsibilities within the work area
- KU4.** Escalation matrix and procedures for reporting work and employment related issues
- KU5.** Various categories of people that one is required to communicate and co ordinate with in the organization
- KU6.** Importance of effective communication in the workplace
- KU7.** Importance of teamwork in organizational and individual success
- KU8.** Various components of effective communication



Qualification Pack

- KU9.** Key elements of active listening
- KU10.** Value and importance of active listening and assertive communication
- KU11.** Barriers to effective communication
- KU12.** Importance of tone and pitch in effective communication
- KU13.** Importance of avoiding casual expletives and unpleasant terms while communicating professional circles
- KU14.** How poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
- KU15.** Importance of ethics for professional success
- KU16.** Importance of discipline for professional success
- KU17.** What constitutes disciplined behaviour for a working professional
- KU18.** Common reasons for interpersonal conflict
- KU19.** Importance of developing effective working relationships for professional success
- KU20.** Expressing and addressing grievances appropriately and effectively
- KU21.** Importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Provide clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems



Qualification Pack

- GS16.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS17.** Report problems that you cannot resolve to appropriate authority
- GS18.** Identify cause and effect relations in their area of work
- GS19.** Use cause and effect relations to anticipate potential problems and their solution



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensure appropriate communication with superiors, peers and others as applicable at work place</i>	10	20	-	-
PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	5	5	-	-
PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	5	5	-	-
PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand	-	10	-	-
<i>Demonstrate appropriate behaviour and etiquette at work place</i>	20	50	-	-
PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible	5	5	-	-
PC5. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	5	5	-	-
PC6. Display appropriate communication etiquette while working	-	10	-	-
PC7. Display active listening skills while interacting with others at work	-	10	-	-
PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	5	5	-	-
PC9. Demonstrate responsible and disciplined behaviours at the workplace	5	10	-	-
PC10. Escalate grievances and problems to	-	5	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N2301: Prepare for operation for Hydrogen injection in Blast Furnace

Description

This OS unit is about performing preparatory activities for carrying out various activities w.r.t. operation for Hydrogen injection in Blast Furnace

Scope

The scope covers the following :

- Carry out preparatory activities for operation for Hydrogen injection in Blast Furnace

Elements and Performance Criteria

Carry out preparatory activities for operation for Hydrogen injection in Blast Furnace

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures
- PC2.** ensure availability of sufficient quantity of raw materials required for smooth operations as per the requirement w.r.t. hydrogen injection
- PC3.** ensure proper functioning of hydrogen injection system
- PC4.** ensure hydrogen injection rates permissible as per SOP
- PC5.** ensure that work area is clean and free of any obstacle or encumbrances
- PC6.** ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of Blast Furnace w.r.t. hydrogen injection as per SOP
- PC7.** coordinate with supply chain of Hydrogen for consistent supply including storage facilities and Hydrogen production source

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** Design parameters of Blast Furnace
- KU3.** Design parameters of Hydrogen injection
- KU4.** Raw materials used in the Blast Furnace
- KU5.** Standard Operating Procedures (SOP) recommended by supplier of hydrogen injection system in the Blast Furnace
- KU6.** Design parameters of hydrogen injection system
- KU7.** Operation and control system of hydrogen injection system
- KU8.** Supply chain of hydrogen with storage and transportation of hydrogen
- KU9.** Various parameters like flow, pressure, temperature and other parameters and control system



Qualification Pack

- KU10.** Chemical reactions in the Blast furnace
- KU11.** Mass balance in the Blast furnace
- KU12.** thermal balance in the Blast furnace
- KU13.** Coke/pulverized coal ratio with hydrogen injection
- KU14.** Process variations with various levels of hydrogen injection
- KU15.** Optimization level of hydrogen injection
- KU16.** Balancing hydrogen with other fuels
- KU17.** Reduction efficiency with hydrogen injection
- KU18.** Real time monitoring system
- KU19.** Reduction in carbon emission with various levels of hydrogen injection
- KU20.** Data analytics

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the team
- GS3.** attentively listen and comprehend the information given by the team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the HOD / incharge or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out preparatory activities for operation for Hydrogen injection in Blast Furnace</i>	30	50	-	20
PC1. obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures	3	6	-	3
PC2. ensure availability of sufficient quantity of raw materials required for smooth operations as per the requirement w.r.t. hydrogen injection	3	6	-	2
PC3. ensure proper functioning of hydrogen injection system	6	9	-	4
PC4. ensure hydrogen injection rates permissible as per SOP	5	8	-	3
PC5. ensure that work area is clean and free of any obstacle or encumbrances	3	6	-	2
PC6. ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of Blast Furnace w.r.t. hydrogen injection as per SOP	4	7	-	3
PC7. coordinate with supply chain of Hydrogen for consistent supply including storage facilities and Hydrogen production source	6	8	-	3
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N2301
NOS Name	Prepare for operation for Hydrogen injection in Blast Furnace
Sector	Iron and Steel
Sub-Sector	
Occupation	Blast Furnace Operation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N2302: Perform operation for Hydrogen injection in Blast Furnace through Tuyeres

Description

This OS unit is about performing operation for Hydrogen injection in Blast Furnace through tuyeres and the standards specified by the organization.

Scope

The scope covers the following :

- Perform operation for Hydrogen injection in Blast Furnace through tuyeres
- Perform post-operation activities

Elements and Performance Criteria

Perform operations with respect to hydrogen injection in Blast Furnace through tuyeres

To be competent, the user/individual on the job must be able to:

- PC1.** use appropriate Personal Protective Equipment (PPE) for safe working in work area
- PC2.** raw materials required for Blast Furnace operation
- PC3.** Feeding system of raw materials
- PC4.** Coordinate with hydrogen supply chain system for required quantity of hydrogen on continuous basis
- PC5.** Monitor real time data to maintain the flow, pressure and other required parameters of hydrogen during process
- PC6.** control the hydrogen flow as per the requirement from time to time
- PC7.** Detect system anomalies like pressure and flow variation
- PC8.** Resolve system anomalies as they occur
- PC9.** Identify hydrogen leakage in the system
- PC10.** Mitigate risks associated with the leakage and other safety issues
- PC11.** Response to emergency situations
- PC12.** Record operational data
- PC13.** Record any deviations in operational issues
- PC14.** Collaborate with operational team for any operational issues
- PC15.** Identify maintenance needs in hydrogen injection system
- PC16.** Collaborate with maintenance team for maintenance work
- PC17.** monitor the CO₂ emission with varying hydrogen flow rates
- PC18.** Establish relation of hydrogen injection with CO₂ emission at different flow rates of hydrogen and savings in CO₂ emission w.r.t. hydrogen injection.

Perform post-operation activities

To be competent, the user/individual on the job must be able to:

- PC19.** update and maintain shift log of production and prepare production and other reports



Qualification Pack

- PC20.** record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book
- PC21.** inform concerned authority in case of any abnormalities observed during execution of work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** The design aspects of hydrogen injection.
- KU3.** Monitoring and control mechanism of hydrogen injection.
- KU4.** Standard Operating Procedures (SOP) recommended by the company
- KU5.** Requirement of raw materials
- KU6.** Utilization efficiency of hydrogen injected
- KU7.** Impact of quality of raw materials on operational practice
- KU8.** Properties of hydrogen including calorific value
- KU9.** Maximum, minimum and optimum flow rates of hydrogen
- KU10.** Maximum, minimum and optimum pressure of hydrogen
- KU11.** Iron ore reduction reactions
- KU12.** Blast Furnace gas composition and utilization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate the process requirements to the team members
- GS2.** write work related information in English/regional language
- GS3.** recognize a workplace problem and take suitable action
- GS4.** plan and organize work according to the work requirements
- GS5.** complete the assigned tasks with minimum supervision
- GS6.** report to the HOD/Section incharge or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform operations with respect to hydrogen injection in Blast Furnace through tuyeres</i>	25	43	-	18
PC1. use appropriate Personal Protective Equipment (PPE) for safe working in work area	1	2	-	1
PC2. raw materials required for Blast Furnace operation	1	1	-	1
PC3. Feeding system of raw materials	1	2	-	1
PC4. Coordinate with hydrogen supply chain system for required quantity of hydrogen on continuous basis	2	4	-	1
PC5. Monitor real time data to maintain the flow, pressure and other required parameters of hydrogen during process	2	4	-	1
PC6. control the hydrogen flow as per the requirement from time to time	2	3	-	1
PC7. Detect system anomalies like pressure and flow variation	2	4	-	1
PC8. Resolve system anomalies as they occur	2	3	-	1
PC9. Identify hydrogen leakage in the system	1	3	-	1
PC10. Mitigate risks associated with the leakage and other safety issues	1	3	-	1
PC11. Response to emergency situations	2	2	-	1
PC12. Record operational data	1	3	-	1
PC13. Record any deviations in operational issues	1	3	-	1
PC14. Collaborate with operational team for any operational issues	1	2	-	1
PC15. Identify maintenance needs in hydrogen injection system	1	2	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Collaborate with maintenance team for maintenance work	1	2	-	1
PC17. monitor the CO2 emission with varying hydrogen flow rates	1	-	-	1
PC18. Establish relation of hydrogen injection with CO2 emission at different flow rates of hydrogen and savings in CO2 emission w.r.t. hydrogen injection.	2	-	-	1
<i>Perform post-operation activities</i>	5	7	-	2
PC19. update and maintain shift log of production and prepare production and other reports	2	3	-	1
PC20. record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book	2	3	-	1
PC21. inform concerned authority in case of any abnormalities observed during execution of work	1	1	-	-
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N2302
NOS Name	Perform operation for Hydrogen injection in Blast Furnace through Tuyeres
Sector	Iron and Steel
Sub-Sector	
Occupation	Blast Furnace Operation
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1704: Apply Decarbonization practice in plant

Description

This OS unit is about apply decarbonization practices in steel plant to reduce carbon emissions.

Scope

The scope covers the following :

- Apply decarbonization practices

Elements and Performance Criteria

Apply decarbonization practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production
- PC2.** calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life)
- PC3.** implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant
- PC4.** implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process
- PC5.** replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.
- PC6.** use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO₂ emissions
- PC7.** use equipment that are powered by electricity instead of fossil fuels
- PC8.** reuse or recycle scrap and waste generated during steelmaking process
- PC9.** switch to renewable electricity and electrifying their vehicle fleets
- PC10.** increase the usage of recycled materials
- PC11.** avoid single use plastics and paper products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** various fuels such as oil, coal etc. responsible for carbon emission in steel plant
- KU3.** various processes such as BOF, DRI, EAF, rotary kiln etc. responsible for carbon emission in steel plant
- KU4.** gases generated in carbon emission i.e. carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) etc.
- KU5.** CO₂ emission and its effect on environment



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- KU6.** Five R's refuse, reduce, reuse, rot, recycle
- KU7.** Carbon capture, usage and storage (CCUS) technology
- KU8.** carbon dioxide removal (CDR) or negative emissions techniques (NETs)
- KU9.** how to calculate CO2 using different fuels
- KU10.** how to control burning of fuel
- KU11.** renewable energy sources such as wind, solar, nuclear, and water

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply decarbonization practices</i>	30	50	-	20
PC1. identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production	3	5	-	2
PC2. calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life)	3	5	-	2
PC3. implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant	3	5	-	2
PC4. implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process	3	5	-	2
PC5. replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.	3	5	-	2
PC6. use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO2 emissions	3	5	-	2
PC7. use equipment that are powered by electricity instead of fossil fuels	3	5	-	2
PC8. reuse or recycle scrap and waste generated during steelmaking process	3	5	-	2
PC9. switch to renewable electricity and electrifying their vehicle fleets	2	4	-	2
PC10. increase the usage of recycled materials	2	3	-	1
PC11. avoid single use plastics and paper products	2	3	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1704
NOS Name	Apply Decarbonization practice in plant
Sector	Iron and Steel
Sub-Sector	
Occupation	Generic
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1703: Apply statistical process control (SPC) techniques

Description

This OS unit is about apply statistical process control (SPC) technique to improve the product quality and production efficiency.

Scope

The scope covers the following :

- Apply statistical process control (SPC) technique

Elements and Performance Criteria

Apply statistical process control (SPC) techniques

To be competent, the user/individual on the job must be able to:

- PC1.** identify the processes that impacts the output of the product
- PC2.** identify the attributes that need to be measured during production
- PC3.** develop a subgroup strategy and sampling plan based on the product's criticality
- PC4.** collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size
- PC5.** calculate standard deviation and variance of the production
- PC6.** prepare normal distribution curve on the basis of data collected
- PC7.** calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes
- PC8.** calculate process capabilities by applying standard formula
- PC9.** calculate process capability index and draw graph on the basis of it
- PC10.** identify improvements required in process on the basis of observations of process capability index results
- PC11.** monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** six sigma concept
- KU2.** statistical process control (SPC) technique and its importance
- KU3.** SPC concept i.e. mode, mean, median, mad
- KU4.** benefits of SPC
- KU5.** objectives of SPC
- KU6.** various tools of SPC
- KU7.** how to perform SPC



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- KU8.** standard deviation and variance
- KU9.** how to prepare normal distribution curve
- KU10.** Process capability index and its calculation formula
- KU11.** how to prepare control charts
- KU12.** concept of accuracy and precision

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply statistical process control (SPC) techniques</i>	30	50	-	20
PC1. identify the processes that impacts the output of the product	2	3	-	1
PC2. identify the attributes that need to be measured during production	2	3	-	1
PC3. develop a subgroup strategy and sampling plan based on the product's criticality	3	5	-	2
PC4. collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size	3	5	-	2
PC5. calculate standard deviation and variance of the production	3	5	-	2
PC6. prepare normal distribution curve on the basis of data collected	3	5	-	2
PC7. calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes	3	5	-	2
PC8. calculate process capabilities by applying standard formula	3	5	-	2
PC9. calculate process capability index and draw graph on the basis of it	3	5	-	2
PC10. identify improvements required in process on the basis of observations of process capability index results	2	4	-	2
PC11. monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern	3	5	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1703
NOS Name	Apply statistical process control (SPC) techniques
Sector	Iron and Steel
Sub-Sector	
Occupation	Generic
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70



Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	45	105	-	-	150	5
ISC/N0009.Work effectively with others	30	70	-	-	100	5
ISC/N2301.Prepare for operation for Hydrogen injection in Blast Furnace	30	50	-	20	100	30
ISC/N2302.Perform operation for Hydrogen injection in Blast Furnace through Tuyeres	30	50	-	20	100	30
ISC/N1704.Apply Decarbonization practice in plant	30	50	-	20	100	10
ISC/N1703.Apply statistical process control (SPC) techniques	30	50	-	20	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	215	405	-	80	700	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.